

20-second cadence of the totally eclipsing white dwarf + M dwarf binary WDJ111803.09–544218.04 (TIC 450781262, $P = 2.25$ h)

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Cadence: 20 s. Sectors: 111–114. Targets: 1.

1 DDT Justification

TIC 450781262 (Gaia DR3 5346312514819760896; $G = 17.38$, $T = 17.41$) was identified on 2026 September 29 as a totally eclipsing post-common-envelope binary with $P = 0.093889794(2)$ d, after the Cycle 9 GI deadline. It falls in Sectors 111–114, before any Cycle 10 observations. No period, eclipse or binary classification is published for it to our knowledge (VSX, SIMBAD, MWDD, VizieR, ADS full text). Earlier 2-min (S90, S99) and 20-s (S90) data come from GI white-dwarf surveys (G07092; G08002/G08062). The target may therefore again be on 2-min lists in S111–S114, which have not yet been released. We request **20-s cadence only**, which is what resolves the white dwarf’s ingress and egress (Sect. 3).

2 Scientific Justification

The system is a low-mass DA white dwarf (MWDD fit 14.9 kK, $\log g$ 7.05, $0.25 M_{\odot}$) with an M4–M5 companion (AllWISE $W1 = 15.52$, 2MASS $K_s = 16.08$ at ~ 300 pc). ATLAS forced photometry (2015–2026) shows a red-dominated reflection modulation (o 8.3%, c 4.6%) and a flat-bottomed eclipse at reflection phase 0.5 that hides the hot component (c -band depth $102 \pm 2\%$; first-to-last contact 7.6 ± 0.2 min; mid-eclipse BJD_{TDB} 2460684.83498). Eclipsing white dwarf + M dwarf binaries with resolved ingress/egress give model-independent white-dwarf radii (e.g. Parsons et al. 2017), and $\sim 0.25 M_{\odot}$ He-core white dwarfs are scarce among them. For $M_2 = 0.15$ – $0.20 M_{\odot}$ ($a = 0.64$ – $0.67 R_{\odot}$) the ingress lasts $\tau = 70$ – 105 s for $R_{WD} = 0.018$ – $0.026 R_{\odot}$, shorter than one 2-min exposure. The total duration T_{14} constrains the companion radius and inclination jointly, and many eclipses give mid-eclipse times for a period-change baseline with ATLAS and TESS FFI data (S10, S37, S63, S64) plus S90 and S99. Masses require radial velocities, which these data do not provide. The product is a photometric radius constraint and an ephemeris. Ground-based high-speed photometry measures single eclipses more precisely; TESS adds ~ 1100 consecutive eclipses.

3 Target Justification and Feasibility

tess-point (0.9.5.post1; Year 9 pointings match tess.mit.edu) places the target on silicon in S111 (cam 4/CCD 3), S112 (4/2), S113 (3/2) and S114 (2/1). We folded the archival SPOC SAP light curves on the ATLAS ephemeris and fitted a trapezoid with exposure smearing (white-noise covariance errors). The results are:

- S90 20 s: depth $0.93 \pm 0.04\%$ of aperture flux, $T_{14} = 394 \pm 19$ s, $\tau = \mathbf{79 \pm 17}$ s
- S90 2 min: $T_{14} = 422 \pm 42$ s, $\tau = 91 \pm 44$ s
- S99 2 min: $T_{14} = 442 \pm 32$ s, $\tau = 126 \pm 34$ s

Mid-eclipse is measured to 4–5 s per sector at either cadence; ingress is constrained only at 20 s. Four further 20-s sectors would reduce σ_τ to ≈ 8 s ($\sqrt{5}$ scaling), $\sim 10\%$ on R_{WD} at fixed geometry. Individual eclipses are not detected (S/N ~ 1 each), so the result relies on stacking. The field is crowded ($b = +5.7^\circ$; CROWDSAP 0.008–0.021; Gaia neighbours $G = 19.6$ at 2.6 arcsec, 18.7 at 5.7 arcsec). Dilution scales the depth only: timing, T_{14} and τ come from the eclipse shape. Contaminating flux sets the noise (about $2\times$ the target flux per 20-s point). No sources brighter than $T = 7$.

4 References

Burke C. J. et al. 2020, ASCL 2003.001 (tess-point); Gaia Collaboration 2023, A&A 674, A1; Gentile Fusillo N. P. et al. 2021, MNRAS 508, 3877; Parsons S. G. et al. 2017, MNRAS 470, 4473; Tonry J. L. et al. 2018, PASP 130, 064505.